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Bandits

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AI and IT in INdustry

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Learning objectives

1. How data-driven marketing works with bandits.
2. How to estimating basic probabilities.
3. Thompson sampling and how it helps to maximize rewards when dealing with uncertain options.

Motivation

p^C

p^S

p^M





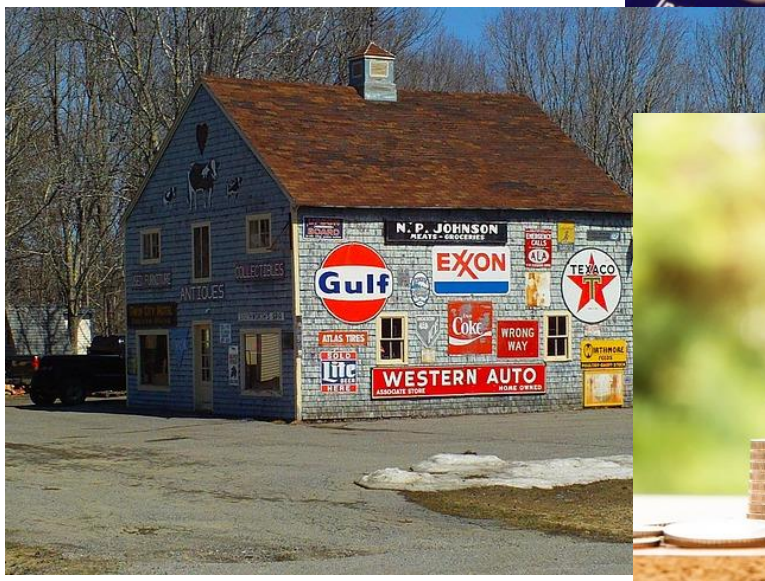
1. How do we estimate p^C, p^S, and p^M?

2. How do we **<u>efficiently</u>** find out which of these three is the largest?

Applications

- Content recommendation in social media.
- Ad selection based on customer profiles.
- Robotics.
- Resource allocation and investment.







Challenges

p^C

p^S

p^M



1. Showing an ad can be expensive for the seller.
2. Neither the seller **nor the customer** have prior information about p^C, p^S, and p^M.

The solution: The UCB algorithm (Upper confidence bound)

Algorithm 1 Upper Confidence Bound Algorithm

Require: $\alpha > 2, T \in \mathbb{N}$

$S_{0,i} = 0, \hat{\mu}_{0,i} = 0,\ i = 1, \ldots, d$

for $t = 1$ **to** T **do**

Choose $A_t = \text{argmin}_{i=1,\ldots,d} \begin{cases} \mu_{t-1,i} - \sqrt{\frac{2\alpha \ln t}{S_{t-1,i}}}, & \text{if } S_{t-1,i} \neq 0 \\ -\infty, & \text{otherwise} \end{cases}$

Observe g_{t,A_t} and pay it

$S_{t,i} = S_{t-1,i} + \mathbf{1}[A_t = i]$

$\hat{\mu}_{t,i} = \frac{1}{S_{t,i}} \sum_{j=1}^{t} g_{t,A_t} \mathbf{1}[A_t = i],\ i = 1, \ldots, d$

end for

Source: https://link.springer.com/article/10.1023/A:1013689704352

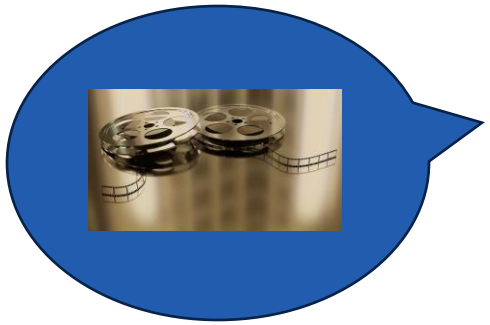
Agenda

- Estimating one probability
  - Customers as urns
- Dealing with multiple probabilities and multiple urns.
  - Proto-UCB sampling
- Dealing with too many customers and too many items.
  - Contextual urns.

Act I: Estimating one probability.

p^M







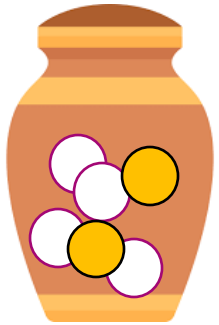
The customer as an urn

p^M





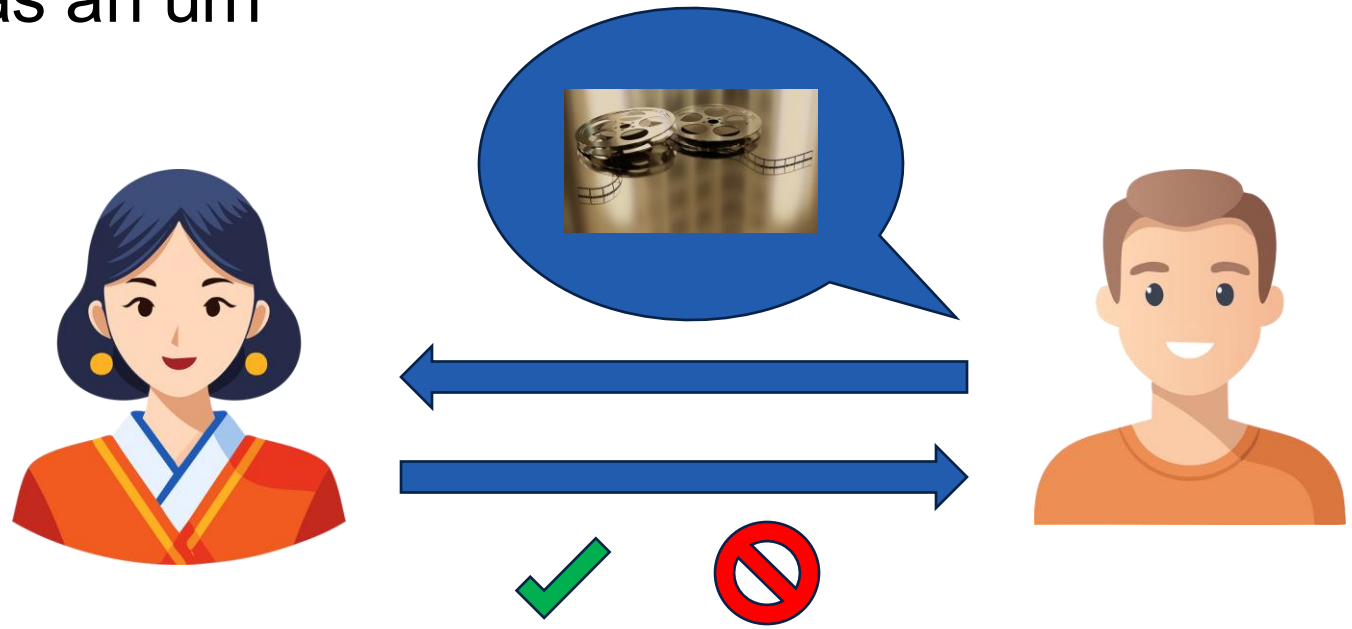
p^M



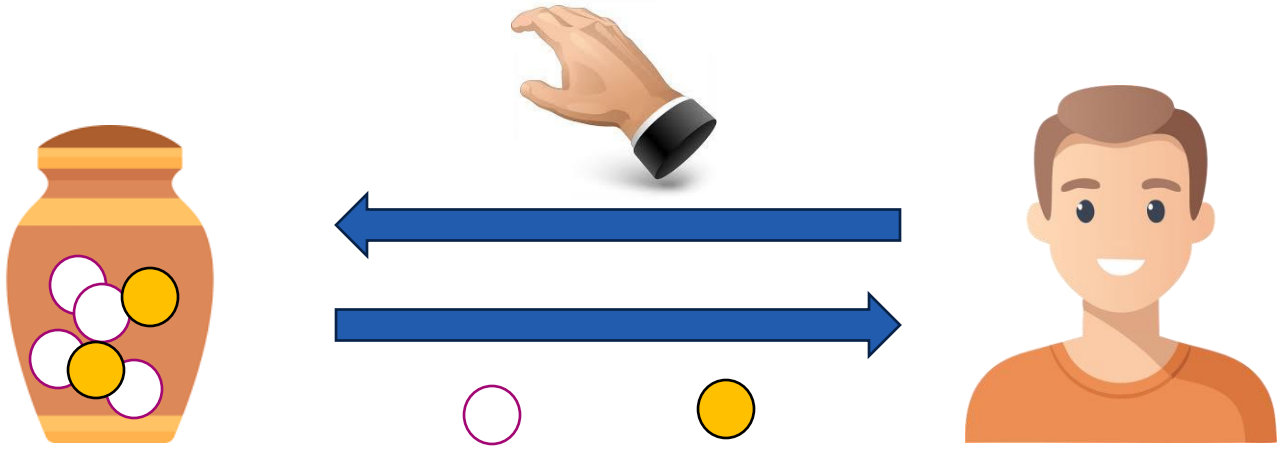


The customer as an urn

$$p^M$$

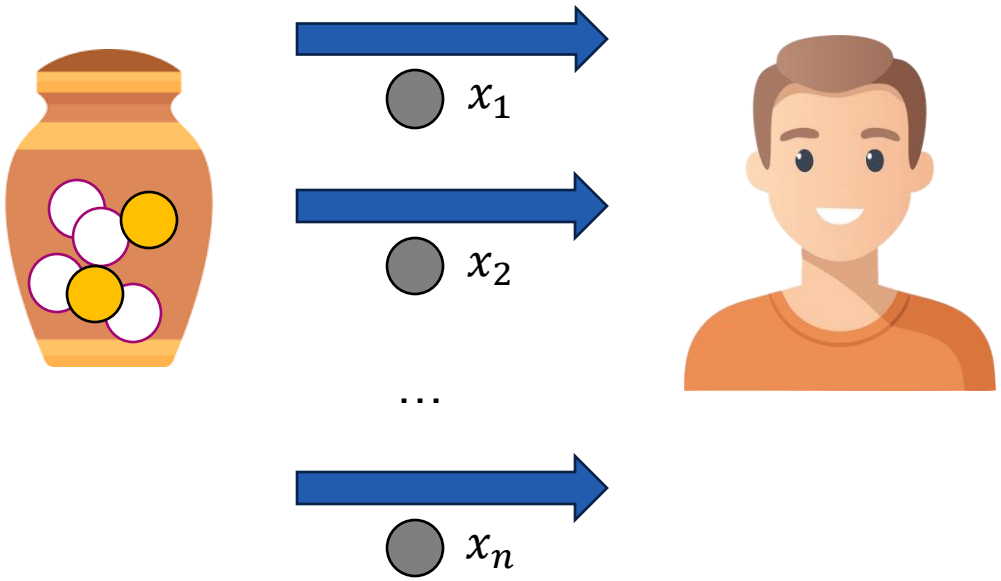


$$p^M = \frac{w^M}{n^M}$$



The empirical estimator

$$p^M = \frac{w^M}{n^M}$$



$$x_i \begin{cases} 1 & if\ ball\ is\ white \\ 0 & if\ ball\ is\ black \end{cases}$$

$$\hat{p}_n{}^M = \bar{x}_n = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

The empirical estimator converges in probability to the true value!

- As n grows $\widehat{p_n}^M$ gets closer and closer to p^M with high probability.

$$\hat{p}_n{}^M \xrightarrow[n\to\infty]{\mathbb{P}} p^M$$

- Formally, for $\delta, \epsilon > 0$, there is $N \in \mathbb{N}$ such that $\mathbb{P}\left(\left|\hat{p}_n{}^M - p^M\right| > \delta\right) < \epsilon$.

- This is an instance of a popular law in probability called **the law of large numbers.**

Confidence intervals

- It is risky to use the empirical estimator when you have only seen a few balls!
- Better use an interval instead of a single estimate!
- A (simple) option is

$$\left(\hat{p}_n - \frac{1}{\sqrt{n}}, \hat{p}_n + \frac{1}{\sqrt{n}}\right).$$



● x 3 ○ x 2 $\left(\frac{2}{5} - \frac{1}{\sqrt{5}}, \frac{2}{5} + \frac{1}{\sqrt{5}}\right)$



● x 38 ○ x 12 $\left(\frac{12}{50} - \frac{1}{\sqrt{50}}, \frac{12}{50} + \frac{1}{\sqrt{50}}\right)$



● x 376 ○ x 124 $\left(\frac{124}{500} - \frac{1}{\sqrt{500}}, \frac{124}{500} + \frac{1}{\sqrt{500}}\right)$

Agenda

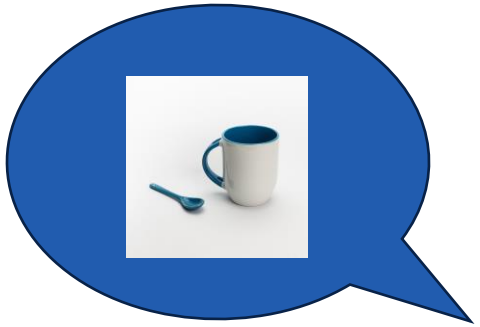
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Act II: Two urns

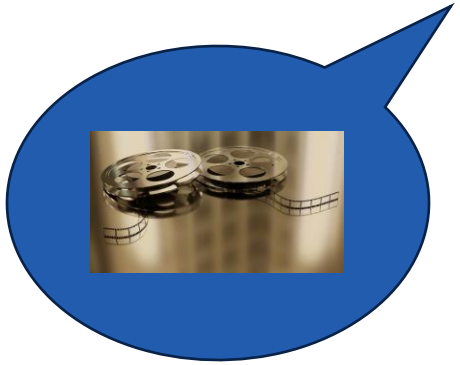
p^C

p^M



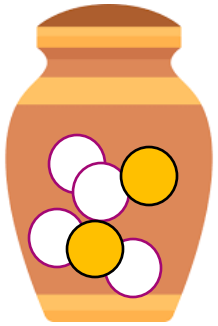




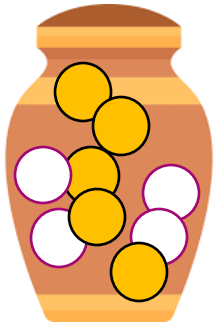


Naïve idea: estimate-then-exploit

p^C



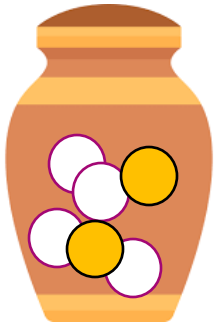
p^M



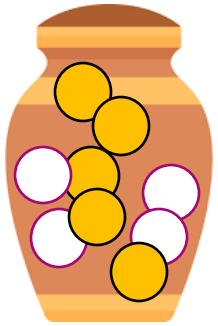


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p^M

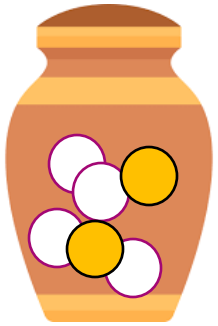




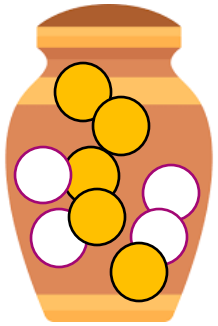
1. Choose a number n.
2. Compute estimators $\hat{p}_n{}^C$ and $\hat{p}_n{}^M$.
3. Exploit the urn with the largest estimate!

Drawbacks!

p^C



p^M



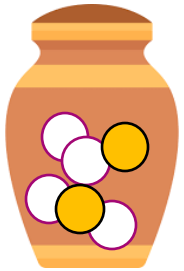


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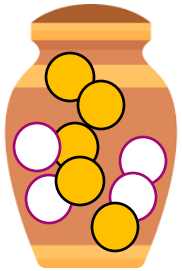
- If p^C and p^M are close and you choose n not large enough, you risk exploiting the wrong urn!
- If p^C and p^M are far apart and you choose n too large, you risk wasting resources estimating the suboptimal urn!

Proto-Thompson sampling!

p^C

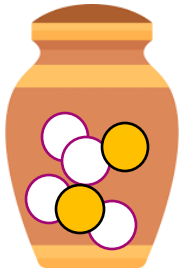


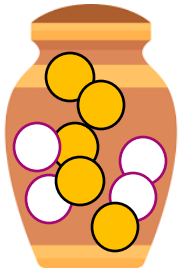
p^M





Proto-UCB sampling!

p^C 

p^M 



$[0, 1]$

$[0, 1]$

1. For each urn u, define the trivial confidence interval $c_u = [0, 1]$.
2. For $t = 1,2, \ldots$

    2.1. Find the urn $\hat{u}$ with the **highest upper bound in the conf interval**.
    2.2. Draw a ball x from $\hat{u}$.
    2.4. **If $x = 1$ get reward :-)**
    2.4. Recompute the confidence interval of $\hat{u}$.

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Act III: Many customers and many items (that is, many urns)

1. Every item gets a representation vector z (e.g. through an autoencoder)
2. For every customer u, we define an unknown representation vector z_u.
3. This leads to an urn with $p(z, z_u) = \max(0, z^\top z_u)$.
4. The task is then to estimate for each customer u its representation vector. This can be done using regression techniques.

What we learned